Positive Square Energy of Every Tricyclic Cactus

25 July 2026

Abstract

We prove that every connected tricyclic cactus G of order n satisfies $s^+(G) \geq n$. The exact block-additive doubly nonnegative bound leaves only the cycle triples $\{3, 3, q\}$ with q odd and $\{3, 5, 5\}$. The first family is settled by packing-two phase, induced cyclic territories, and a product-subpartition comparison. For $\{3, 5, 5\}$, a bridge separates a one-cycle packet unless all cycles form one shared-cut cluster; the latter has four cores, handled by an exact positive-coefficient phase certificate. Arbitrary bridge blocks and arbitrary attached trees are allowed.

1 Spectral and matching preliminaries

All graphs are finite, simple, and undirected. For the adjacency eigenvalues $\lambda_1, \ldots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \quad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2, \quad D(G) = s^+(G) - s^-(G).$$

Since a connected tricyclic graph has $|E(G)| = n + 2$,

$$s^+(G) + s^-(G) = 2n + 4, \qquad s^+(G) = n + 2 + \frac{1}{2} D(G). \tag{1}$$

These square energies originate in spectral coloring bounds [1]. Liu–Tang–Zhang proved the general $n-1$ conjecture [4]; the stronger edge-surplus conjecture of Akbari–Kumar–Mohar–Pragada–Zhang asks for $s^+(G) \geq n$ when $|E(G)| \geq n + 1$ [3]. Ning–Zeng proved the odd-unicyclic sign theorem used below in its one-cycle form [5].

For positive activities $a = (a_v)$ define

$$Z_F(a) = \sum_{M \text{ matching of } F} \prod_{v \text{ unmatched by } M} a_v, \qquad Z_F(t) = Z_F((t)_v),$$

and normalize the characteristic polynomial by

$$\Psi_G(t) = i^{-n} \det(itI - A(G)) = \prod_j (t + i\lambda_j).$$

Sachs' formula, grouped by collections $\mathcal{C}$ of pairwise vertex-disjoint cycles, is

$$\Psi_G(t) = \sum_{\mathcal{C}} \prod_{C \in \mathcal{C}} (-2i^{-|C|}) Z_{G - V(\mathcal{C})}(t). \tag{2}$$

Thus a $3 \pmod 4$ cycle has multiplier $-2i$, and a $1 \pmod 4$ cycle has multiplier $2i$. If $\Theta_G(t)$ is the continuous argument of $\Psi_G(t)$ tending to zero at infinity, then

$$D(G) = -\frac{4}{\pi} \int_0^\infty t\Theta_G(t)\, dt. \tag{3}$$

Indeed, integrate $\lambda|\lambda| = (2/\pi)\int_0^\infty \lambda^3/(\lambda^2+t^2)\,dt$, sum over the spectrum, use $\sum \lambda_j = 0$, and integrate by parts. This also fixes the sign and branch in (3).

We use two standard exact reductions. First, for every vertex partition (V_j), induced-subgraph superadditivity gives

$$s^+(G) \geq \sum_j s^+(G[V_j]) \tag{4}$$

[2]. Second, a tree branch oriented toward its attachment satisfies

$$Q_{u\to p} = \frac{Z_{T_{u\to p}}(t)}{Z_{T_{u\to p}-u}(t)} = t + \sum_{w \text{ child of } u} Q_{w\to u}^{-1} \geq t. \tag{5}$$

Eliminating all branches therefore gives a common positive factor $K(t)$ and activity $a_v = t + y_v$, $y_v \geq 0$, at every retained vertex, in every Sachs term. This follows by splitting a matching according as its root is unmatched or matched to one child.

Lemma 1.1 (Phase and packet bounds)**.** *The following hold, with arbitrary tree attachments.*

1. *If the graph contains a cycle, every cycle is* $3 \pmod 4$*, and at most two are pairwise vertex-disjoint, then* $D > 0$*.*

2. *A unicyclic* C_q *cactus with* $q = 1 \pmod 4$ *satisfies* $s^+ \geq h - \delta_q$*, where* $\delta_q = \sec(\pi/q) - 1$*.*

3. *A bicyclic cactus with blocks* C_3, C_q*,* $q = 1 \pmod 4$*, satisfies* $s^+ > h + 1 - \delta_q$*. Every bicyclic cactus satisfies* $s^+ \geq h$*.*

Proof. In (1), (2) has strictly negative imaginary part: singleton cycle terms are negative imaginary, while empty and two-cycle terms are real. Thus $\Theta < 0$, and (3) gives $D > 0$. This is the packing-two theorem, including the unicyclic result of Ning–Zeng.

For (2), (5) gives $\Psi/K = Z_{C_q}(a) + 2i$. Since $a_v \geq t$, coefficientwise monotonicity of Z bounds its phase by that of the bare cycle. The eigenvalues $2\cos(2\pi j/q)$ give

$$s^+(C_q) = q + 1 - \sec(\pi/q), \qquad D(C_q) = -2\delta_q. \tag{6}$$

Coulson integration proves (2).

For the mixed packet, retain the two cycles and their joining path. Put $A = Z_{H-V(C_3)}(a)$, $B = Z_{H-V(C_q)}(a)$, and let D_0 be the partition after deleting both cycles when they are disjoint. Sachs gives

$$\Psi/K = R + 2i(B - A), \quad R = Z_H(a) \quad \text{or} \quad Z_H(a) + 4D_0.$$

Partitioning matchings by whether they use a boundary edge of C_q gives $Z_H = Z_{C_q}(a|_{C_q})B + E$, $E \geq 0$. Write $Z_{C_q}(a|_{C_q}) = Z_{C_q}(t) + L$, $L \geq 0$. Then

$$R - Z_{C_q}(t)(B - A) = E + LB + Z_{C_q}(t)A \ (+4D_0) > 0.$$

Hence $\Theta < \operatorname{Arg}(Z_{C_q}(t) + 2i)$, so $D > D(C_q)$ and (1) in the bicyclic form proves (3)'s mixed bound. The final assertion is the established bicyclic-cactus theorem [6]; it follows there from the same sharp DNN reduction and the two residual phase arguments. These proofs use neither triangle-ear nor edge-monotonicity. $\square$

2 The exact DNN reduction

For a connected edge-containing graph define

$$\kappa(G)=\sup_{\substack{M\succeq 0,\ M\geq 0\\ \mathbf{1}^TM\mathbf{1}>0}}\frac{4(\sum_{uv\in E(G)}\sqrt{M_{uv}})^2}{\mathbf{1}^TM\mathbf{1}}.$$

Lemma 2.1 (Sharp cactus DNN constant)**.** *If a cactus has b bridge blocks and cyclic blocks of lengths $\ell_1,\dots,\ell_r$, then*

$$\kappa(G)=b+\sum_j\kappa(C_{\ell_j}),\qquad \kappa(C_\ell)=\begin{cases}\ell, & \ell \text{ even},\\ \ell\sec^2(\pi/(2\ell)), & \ell \text{ odd},\end{cases}$$

and $s^-(G)\leq\kappa(G)$.

Proof. Fold every nonedge entry M_{uv} into the two diagonals by adding $M_{uv}(e_u-e_v)(e_u-e_v)^T$. The denominator and edge entries are unchanged. On C_ℓ, rotational averaging and concavity of the square root reduce to common edge entry x and row sum r. A Fourier eigenvalue gives $r\geq 4x$ for even ℓ and $r\geq 2(1+\cos(\pi/\ell))x$ for odd ℓ. Equality is attained by $x(2cI+A(C_\ell))$, with $c=1$ or $c=\cos(\pi/\ell)$, proving the formula. Also $\kappa(K_2)=1$.

For explicit cut-vector bookkeeping, suppose $G=G_1\vee_z G_2$. In a folded Gram system write $g_z=u+w+h$, where u lies in the span U of side 1, w lies in the orthogonal span W of side 2, and $h\perp U+W$. Use $u+h$ as the cut vector on side 1 and w on side 2. Edge inner products remain unchanged and, if a,b are the sums of the noncut vectors on the two sides,

$$\|a+b+u+w+h\|^2=\|a+u+h\|^2+\|b+w\|^2.$$

Thus both the edge numerator sum and the denominator split. Cauchy–Schwarz gives $\kappa(G)\leq\kappa(G_1)+\kappa(G_2)$. Conversely, glue near-optimal Gram systems orthogonally and scale them so $T_1/\kappa(G_1)=T_2/\kappa(G_2)$; this gives equality. Iteration proves block additivity.

Finally write $A=P-B$ for the positive and negative spectral parts. The Schur product theorem makes $M=B\circ B$ DNN, while $s^-/2=-\sum_{uv\in E}B_{uv}\leq\sum_{uv\in E}|B_{uv}|$ and $\mathbf{1}^TM\mathbf{1}=\operatorname{tr}B^2=s^-$. The definition of κ gives $(s^-)^2\leq\kappa(G)s^-$, proving the spectral bound. □

Put $\epsilon_\ell=0$ for even ℓ and $\epsilon_\ell=\ell\tan^2(\pi/(2\ell))$ for odd ℓ. Then

$$\epsilon_3=1,\quad \epsilon_5=5-2\sqrt{5},\quad \epsilon_5+\epsilon_7<1,\tag{7}$$

and ϵ_ℓ strictly decreases through odd integers: after setting $u=\pi/(2x)$ this is the monotonicity of $\tan^2 u/u$, whose derivative is positive because $2u>\sin u\cos u$. The last inequality follows, for example, from $\cos(\pi/7)>7/8$.

If the tricyclic cycle lengths are p,q,r, then $b+p+q+r=n+2$. Lemma 2.1 yields

$$s^+(G)\geq n+2-\epsilon_p-\epsilon_q-\epsilon_r.\tag{8}$$

Thus DNN proves $s^+\geq n$ unless the excess sum exceeds 2. Using (7) and monotonicity, this happens for exactly

$$\{3,3,q\}\ (q\geq 3\text{ odd}),\qquad \{3,5,5\}.\tag{9}$$

Indeed an even length contributes zero; without two triangles the only possible maximum is $\epsilon_3+2\epsilon_5>2$, while replacing either five by at least seven makes the sum <2.

3 Two triangles and one odd cycle

Proposition 3.1. *A cactus whose cyclic blocks are C_3, C_3, C_q, q odd, satisfies $s^+ > n$.*

Proof. First let $q \equiv 3 \pmod 4$; this includes the all-triangular case $q = 3$. If the cycle-packing number is at most two, Lemma 1.1(1) gives $D > 0$ and hence $s^+ > n+2$. If the packing number is three, the cycles are vertex-disjoint. Cut bridge edges in the minimal block tree joining them and assign every hanging tree to its attachment. This partitions $V(G)$ into three connected induced unicyclic territories. Each has $D > 0$, hence s^+ greater than its order; (4) gives $s^+(G) > n$.

Now let $q \equiv 1 \pmod 4$ and $\delta_q = \sec(\pi/q) - 1 < 1$. If all cycles are disjoint, choose an adjacent pair in the reduced three-node cycle tree. Two triangles form a packet with $s^+ > h+1$, leaving a C_q packet with $s^+ \geq h - \delta_q$; or a mixed pair has $s^+ > h + 1 - \delta_q$ and leaves a triangular packet with $s^+ > h$. Thus (4) gives $s^+ > n + 1 - \delta_q > n$. The same bridge cut works when one triangle is disjoint from the other triangle and C_q.

Suppose next that the triangles are disjoint and C_q meets both, at distinct cut vertices x_1, x_2. Give x_1, its triangle, and branches off the long-cycle side to one induced territory. The complement is connected because $C_q - x_1$ is a path. Both territories are triangular unicyclic cacti, so again $s^+ > n$ by (4).

It remains that the triangles T_1, T_2 share a cut vertex. Retain the cyclic core and joining block paths and apply (5). For a cycle collection J, put $D_J = Z_{H-V(J)}(a)$ when its cycles are disjoint and $D_J = 0$ otherwise. With $B = D_Q$ and $A_j = D_{T_j}$, Sachs gives

$$\Psi_G/K = R + iI, \quad R = Z_H + 4D_{T_1,Q} + 4D_{T_2,Q}, \quad I = 2(B - A_1 - A_2).$$

There is neither a double-triangle nor a triple term because the triangles meet. Partition matchings by boundary edges of $Q = C_q$:

$$Z_H = Z_Q(a|_Q)B + E, \quad E \geq 0, \qquad Z_Q(a|_Q) = Z_q(t) + L, \quad L \geq 0.$$

Consequently

$$R - \frac{Z_q(t)I}{2} = E + LB + 4D_{T_1,Q} + 4D_{T_2,Q} + Z_q(t)(A_1 + A_2) > 0.$$

Equivalently, $\operatorname{Im}((\Psi_G/K)\overline{(Z_q + 2i)}) < 0$. The continuous arguments tend to zero at infinity, so they cannot cross and $\Theta_G < \operatorname{Arg}(Z_q + 2i)$. Equations (3), (6), and (1) give $s^+(G) > n + 2 - \delta_q > n$.

For exhaustion: either the triangles meet, or they do not. In the latter case Q meets both, at most one, or neither; these are respectively the two-territory case, the separating-bridge case, and the reduced-tree case. Thus every block incidence has been covered. □

4 A triangle and two pentagons

Join two cycle blocks in the *shared-cut graph* when they share a cut vertex. If this graph is disconnected, contract each shared-cut component and all acyclic paths between components. The resulting cluster incidence is a tree. With three cycles, it has a leaf cluster consisting of exactly one cycle: for a $2+1$ split the singleton is a leaf, and for a $1+1+1$ split any leaf is. Cutting the first bridge toward the rest partitions G into two connected induced territories, one containing exactly that cycle.

Proposition 4.1 (Bridge-separable case)**.** *If the shared-cut graph for cycle lengths $\{3, 5, 5\}$ is disconnected, then $s^+(G) > n$.*

Proof. Use the leaf-cluster cut above. If the isolated packet is triangular, its s^+ exceeds its order, while the remaining $\{5, 5\}$ bicyclic packet has s^+ at least its order by Lemma 1.1(3).

If the isolated packet is pentagonal, put $\delta_5 = \sec(\pi/5) - 1 = \sqrt{5} - 2$. Lemma 1.1 gives $s^+(U_5) \geq u - \delta_5$ and $s^+(B_{3,5}) > b + 1 - \delta_5$. Superadditivity therefore gives

$$s^+(G) > n + 1 - 2\delta_5 = n + 5 - 2\sqrt{5} > n.$$

This includes three separate cyclic clusters; the cut is an actual bridge and the two territories are induced, so no edge-monotonicity is invoked. □

Proposition 4.2 (One shared-cut cluster)**.** *If the three blocks C_3, C_5, C_5 form one connected shared-cut cluster, then*

$$s^+(G) > n + 6 - 2\sqrt{5} > n.$$

Proof. There are exactly four incidences up to isomorphism: a three-cycle bouquet; the triangle as middle block with distinct cut vertices; or a pentagon as middle block with its two cut vertices at cyclic distance one or two. These exhaust a connected three-node block tree, including coincident incidences. All bridges are now branches off the cyclic core H, so (5) gives independent positive core activities $a_v \geq t$ and a common positive factor.

Let T, P, Q denote the triangle and pentagons and put $D_J = Z_{H-V(J)}(a)$, interpreted as zero when the cycles in J meet. Sachs gives $\Psi_G/K = R + iI$, where

$$\begin{aligned} R &= Z_H + 4D_{T,P} + 4D_{T,Q} - 4D_{P,Q}, \\ I &= -2D_T + 2D_P + 2D_Q. \end{aligned}$$

There is no triple term. Let $p = Z_P(a|_P)$ and $q = Z_Q(a|_Q)$ be the actual weighted isolated-pentagon partitions. Exact expansion over $\mathbb{Z}[a_v]$ gives the certificate

$$\Phi = 2R(p + q) - I(pq - 4) > 0 \tag{10}$$

for every positive activity vector. Table 1 records the complete coefficient audit; the script constructs all matching partitions recursively, checks the cores, and checks every coefficient.

core	terms	min	max
bouquet	2547	2	32
`8112f2944f9823177afd48deccfb958ac960548e09d9d838e4965c33eb39e979`			
triangle middle	2192	2	32
`4fdb04cee38f0c0e2ac2de6dff7e641c2e190a3c018af8308a5c69e378de2ba2`			
pentagon middle, distance 1	2925	2	36
`bfa73346f169f28ec6109418ce22fe44daaae6b081ade30074932b139f0828f4`			
pentagon middle, distance 2	2895	2	36
`5c6c213471a856cb743afda8e62407d7d27de5984b612c70ab411499011db437`			

Table 1: Exact coefficient certificates for Φ.

Now $(p + 2i)(q + 2i) = (pq - 4) + 2i(p + q)$, and (10) is precisely

$$\operatorname{Im}\left((p + 2i)(q + 2i)\overline{(R + iI)}\right) > 0.$$

The continuous arguments all tend to zero at infinity. Strict positivity prevents a branch crossing, hence

$$\Theta_G(t) < \operatorname{Arg}(p + 2i) + \operatorname{Arg}(q + 2i).$$

Because $a_v \geq t$, each weighted isolated-pentagon phase is at most the bare C_5 phase. Coulson integration and $D(C_5) = 4 - 2\sqrt{5}$ therefore give $D(G) > 2D(C_5) = 8 - 4\sqrt{5}$. Equation (1) proves the claim. □

5 Conclusion

Theorem 5.1. *Every connected tricyclic cactus G on n vertices satisfies*

$$\boxed{s^+(G) \geq n}.$$

Proof. The sharp DNN bound (8) proves the assertion unless the cycle length triple is one of (9). Proposition 3.1 settles every $\{3, 3, q\}$. For $\{3, 5, 5\}$ the shared-cut graph is either disconnected, covered by Proposition 4.1, or connected, covered by Proposition 4.2. These alternatives exhaust all cycle lengths, all block-cut incidences, all bridges, and all attached trees. □

A Exact reproduction and local proof objects

From the repository root run

```
python positive-square-energy/experiments/c3_c5_c5_shared_cluster_certificate.py
```

to reproduce Table 1. The public local manuscripts used for the packet lemmas and audited assembly are

```
all-bicyclic-cacti/paper.tex
packing-two-square-energy/paper.tex
research/c3-c3-cq-tricyclic-cactus-2026-07-25.md
research/c3-c5-c5-shared-cluster-2026-07-25.md
```

References

[1] A. Abiad, L. de Lima, D. N. Desai, K. Guo, L. Hogben, and J. Madrid, *Positive and negative square energies of graphs*, Electron. J. Linear Algebra 39 (2023), 307–326.

[2] S. Akbari, H. Kumar, B. Mohar, and S. Pragada, *A linear lower bound for the square energy of graphs*, Electron. J. Combin. 32(3) (2025), Paper No. P3.53.

[3] S. Akbari, H. Kumar, B. Mohar, S. Pragada, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264, 2025.

[4] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.

[5] B. Ning and J. Zeng, *A proof of a conjecture on positive and negative square energies of unicyclic graphs*, arXiv:2605.24668, 2026.

[6] Companion manuscript, *Positive Square Energy of Every Bicyclic Cactus*, 2026; repository path `all-bicyclic-cacti/paper.tex`.